A Framework for Retrospective Analysis and Case Studies of Internet Crimes Against Children Across U.S. Task Forces, 2010–2026

Mrinaal Ramachandran

Graduate Student, University of Massachusetts Amherst

Independent Research — CaseLinker

ICAC Case Studies — *First Draft of Report #4*

Written April 2026

Abstract

The history of the internet has been largely championed from the perspective of platforms, growth, and innovation. Largely absent from that record is a systematic account of how the same infrastructure transformed the landscape of child exploitation—and how law enforcement institutions adapted, strained, and responded across three decades of technological change. This gap is not incidental: internet-facilitated crimes against children have scaled alongside the web itself, yet their history remains fragmented across jurisdictions, documented in press releases but never systematically connected at scale, and invisible at the institutional level.

This paper presents the first longitudinal case study analysis of the Internet Crimes Against Children (ICAC) Task Force Program across 2010–2026, organized across four technological eras: the social platform era (2010–2014), the mobile-first transition (2015–2018), platform proliferation and expansion (2019–2022), and the social media and generative AI period (2023–2026). Drawing on 2500 publicly available case reports processed through CaseLinker—an open-source retrospective case analysis system—we develop 20 structured case studies that trace how perpetrator methodologies, platform surfaces, and law enforcement responses co-evolved across each era.

The contribution of this paper is both historical and methodological. Historically, the case studies span platform involvement, victim and perpetrator demographics, investigation type (proactive, undercover, and reactive), geographic and jurisdictional patterns, and agency coordination networks. Each era is characterized by shifting perpetrator methodologies, evolving platform surfaces, and corresponding law enforcement adaptations—patterns that are invisible at the individual case level and only emerge through systematic cross-case analysis. Methodologically, CaseLinker provides an auditable, open-source infrastructure for retrospective case analysis, designed to be replicable and extensible by other researchers working in sensitive domains. This paper establishes the framework through which ICAC reports will be sampled, analyzed, and presented along with the ethical, legal, and practical

considerations

Research Methodology

This research is grounded in CaseLinker, an open-source retrospective case analysis system developed to process, structure, and surface patterns from publicly available ICAC press releases and task force case reports. The system is designed to be auditable, interpretable, and appropriate for the sensitivity of the subject matter—operating exclusively on already-public, already-redacted case summaries and producing no outputs that identify individuals.

Case study selection. The 20 case studies will be selected using stratified sampling across the four historical eras and five analytic dimensions—platform, demographics, investigation type, geography, and agency network—to ensure cases appropriately represent the nature of this crime and how its been investigated. Each case study is written as a structured narrative: platform context, perpetrator methodology, investigative approach, prosecutorial outcomes, and relationship to the broader technological moment of its era. Relevant redacted summaries will be obtained through Arizona's Public Records Law (A.R.S. § 39-121) and Georgia's Open Records Act (O.C.G.A. § 50-18-70 et seq.) for cases originating from those jurisdictions, with equivalent statutes applied by state as applicable — including but not limited to Florida (F.S. § 119.01), Texas (Tex. Gov't Code § 552), California (Cal. Gov't Code § 6250 et seq.), New York (N.Y. Pub. Off. Law § 84 et seq.), and Illinois (5 ILCS 140). All requested records are limited to closed, adjudicated cases to minimize exemption risk under applicable state public record laws. The goal is not statistical generalization from the sample, but grounded historical illustration of patterns and insights that high level statistics and that individual case reading cannot reveal. All case data is drawn from public sources. The full source corpus, extraction code, and analysis pipeline are released as open-source software to support replication and extension by other researchers.

Framework for Case Studies Selected case reports and their redcated case summaries will be analyzed from the 5 perspectives listed above: platform context, perpetrator methodology, investigative approach, prosecutorial outcomes, and the case's relationship to 19 other selected case studies. No attempt will be made to re-identify individuals, reconstruct victim or further divulge perpetrator identity, or supplement public records with external investigation. Given the sensitivity of the subject matter, dissemination follows a restricted-first approach: findings are shared initially with practitioner and research contacts prior to public release, and any case detail that risks identification — even in aggregate — is withheld or further abstracted.

CaseLinker and the collection and analysis of ICAC case reports received a determination from the University of Massachusetts Amherst Human Research Protection Office (HRPO Determination #7668); this research does not contain private or identifiable information under federal regulations [45 CFR 46.102(f)(1), (2)]. Relevant research, legal, and practitioner considerations will be taken before beginning case studies.

Initial References - note : preliminary copy of prominent links and papers I liked, not an exhaustive list of all my references

[1] National Center for Missing & Exploited Children. (2024). *2024 CyberTipline Data Report.* https://www.missingkids.org/gethelpnow/cybertipline/cybertiplinedata

[2] Office of Juvenile Justice and Delinquency Prevention. (2024). *Internet Crimes Against Children Task Force Program.* https://ojjdp.ojp.gov/programs/internet-crimes-against-child

[3] Souras, Y. (2025, March 26). *Testimony before the U.S. House Committee on Energy and Commerce regarding CyberTipline reporting and the REPORT Act.* https://www.congress.gov/119/meeting/house/118066/witnesses/HHRG-119-IF17-Wstate-SourasY-20250326.pdf

[4] Internet Watch Foundation. (2024, July). *How AI is being abused to create child sexual abuse imagery: Updated report.* https://www.iwf.org.uk/about-us/why-we-exist/our-research/how-ai-is-being-abused-to-create-child-sexual-abuse-imagery/

[5] eSafety Commissioner. (2026, January 8). *Generative AI and child safety: A convergence of innovation and exploitation.* https://www.esafety.gov.au/newsroom/blogs/generative-ai-and-child-safety-a-convergence-of-innovation-and-exploitation

[6] Technology Coalition. (2023). *An update on voluntary detection of CSAM.* https://technologycoalition.org/resources/update-on-voluntary-detection-of-csam/

[7] Thorn. (2023, September). *Hashing and matching is core to proactive CSAM detection.* https://safer.io/resources/hashing-and-matching-is-core-to-proactive-csam-detection/

[8] Thorn. (2025, January). *Cross-platform CSAM hash sharing for platforms.* https://safer.io/resources/optimize-csam-detection-with-saferlist/

[9] Steinebach, M. (2023). An analysis of PhotoDNA. *ARES 2023* (18th International Conference on Availability, Reliability and Security). https://dl.acm.org/doi/fullHtml/10.1145/3600160.3605048

[10] Deryck, M., Leblanc-Albarel, D., & Preneel, B. (2026). White-box attacks on PhotoDNA perceptual hash function. *IACR ePrint, 2026/486.* https://eprint.iacr.org/2026/486.pdf

[11] United Nations Interregional Crime and Justice Research Institute. (2022). *AI for Safer Children Global Hub.* https://unicri.org/topics/AI-for-Safer-Children

[12] Burns, C. M., Morley, J., Bradshaw, R., & Domene, J. (2008). The emotional impact on and coping strategies employed by police teams investigating internet child exploitation. *Traumatology, 14*(2), 20–31.

[13] Perez, L. M., Jones, J., Englert, D. R., & Sachau, D. (2010). Secondary traumatic stress and burnout among law enforcement investigators exposed to disturbing media images. *Journal of Police and Criminal Psychology, 25*(2), 113–124.

[14] Klein, G. (1998). *Sources of Power: How People Make Decisions.* MIT Press.

[15] ICAC Task Force Public Case Reports — Arizona, Idaho, Texas, Georgia, and additional task forces (2010–2026). Collected and indexed via the CaseLinker public corpus.

[16] Ramachandran, M. (2026). *CaseLinker: An open-source system for cross-case analysis of Internet Crimes Against Children reports.* Technical Report #1. University of Massachusetts Amherst. https://doi.org/10.48550/arXiv.2603.18020

[17] Ramachandran, M. (2026, March). *Why Internet Crimes Against Children And Retrospective Case Analysis Matters* Position Piece. https://mrinaalr.github.io/website/Why%20Internet%20Crimes%20Against%20Children%20And%20Retrospective%20Case%20Analysis%20Matters.pdf